

Date	15 March 2026
Team ID	XXXXXX
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposal report aims to transform credit card approval using machine learning, boosting efficiency, transparency, and accuracy. It tackles current system inefficiencies, promising better operations, reduced biases, and happier customers. Key features include a machine learning-based credit model, real-time decision-making, and explainable AI.

Project Overview	
Objective	The primary objective is to revolutionize the credit card approval process by implementing advanced machine learning classification techniques, ensuring faster, fairer, and more accurate assessments.
Scope	The project comprehensively assesses and enhances the approval pipeline, incorporating ML and alternative data for a more inclusive and efficient system.
Problem Statement	
Description	Addressing the rigidities and "black-box" nature of traditional credit scoring systems that lead to unexplained rejections for applicants with thin credit files.
Impact	Solving these issues will result in improved financial inclusion, reduced manual review bottlenecks, and an overall enhancement in applicant trust and customer satisfaction.
Proposed Solution	
Approach	Employing machine learning algorithms (like XGBoost) paired with SHAP to analyze creditworthiness, creating a dynamic, adaptable, and transparent approval system.

Key Features	• Implementation of an ML-based alternative credit assessment model. • Real-time decision-making for quicker credit card approvals.
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	• Integrated Explainable AI (XAI) to justify predictions and prevent black-box decision making.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard vCPUs (or T4 GPU for heavy training)
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	500 GB - 1 TB SSD
Software		
Frameworks	Python/JS frameworks	FastAPI (Backend), React (Frontend)
Libraries	Additional libraries	scikit-learn, xgboost, shap, pandas, numpy
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle Credit Card Approval dataset, CSV format